

provide *in vivo* insights into human enhancer function but can be limited by throughput (*Nature* **643**, 839–846; 2025).

The generation of large-scale datasets has fueled the development of machine-learning tools to decode regulatory logic and predict regulatory sequences. Computational advances in sequence to function models help to design synthetic enhancers with cell-type specificity (*Nat. Genet.* **54**, 613–624; 2022, *Nature* **626**, 207–211; 2024, and *Nature* **626**, 212–220; 2024). These synthetic enhancers can be more effective in driving cell-type-specific expressions as compared with their natural counterparts (*Nature* **634**, 1211–1220; 2024).

Despite these advances, the spatiotemporal regulation of gene expression by multiple enhancers and other regulatory elements remains poorly understood. Enhancers rarely function in isolation; their activity is influenced by interactions with transcription factors, DNA accessibility, histone modifications and chromatin architecture. To fully elucidate enhancer function, we envision that high-throughput tools operating in endogenous contexts and accounting for chromatin dynamics will be essential. In parallel, genomic language models trained on DNA sequences hold promise for enhancing our understanding of enhancer grammar and guiding the rational design of regulatory elements.

Predicting RNA structures

Predicting the folded structures of RNA molecules poses greater challenges than proteins, but steady progress continues.

By Allison Doerr

The impact that the highly accurate protein-structure prediction tool AlphaFold2 (*Nature* **596**, 583–589; 2021) made on biology research was undeniable and immediate – so much so that protein structure prediction was the obvious choice for our Method of the Year in 2021 (*Nat. Methods* **19**, 1; 2022).

As with proteins, many noncoding RNA molecules such as ribozymes, riboswitches and aptamers also adopt folded tertiary structures



Methods to predict folded RNA structures are advancing.

that are key to carrying out their biological functions. The tertiary interactions responsible for an RNA molecule's fold are relatively weak, however, so a given sequence may fold into different three-dimensional arrangements depending on the binding partner or environment it is found in. Although the experimental techniques commonly used for protein structure determination – X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy and cryo-electron microscopy (cryo-EM) – can also be applied to structured RNA molecules, their conformational flexibility poses substantial challenges for these techniques.

Inspired by AlphaFold's success, some researchers are also advancing machine-learning-based structure-prediction approaches for RNA. A pioneering paper described a method, trained on 18 known RNA structures, that could accurately identify native-like RNA folds (*Science* **373**, 1047–1051; 2021). Several deep-learning-based approaches to predict RNA tertiary structure soon followed, including DeepFoldRNA (*bioRxiv*, <https://doi.org/10.1101/2022.05.15.491755>; 2022), DRfold (*Nat. Commun.* **14**, 5745; 2023), trRosettaRNA (*Nat. Commun.* **14**, 7266; 2023), RhoFold+ (*Nat. Methods* **21**, 2287–2298; 2024) and NuFold (*Nat. Commun.* **16**, 881; 2025).

In the protein-structure prediction field, competitions such as CASP have had a major role in driving the development of new methods – including AlphaFold, which first stunned the other competitors in 2020. The CASP competition now includes an RNA-structure prediction component (*Proteins* **91**, 1747–1770;

2023). RNA-Puzzles is another long-running competition dedicated to advancing RNA structure prediction (*Nat. Methods* **22**, 399–411; 2025).

The biggest hurdle to greater success, however, is the lack of substantial experimental training data for machine-learning algorithms. Just over **2,000 RNA-only structures** are available in the Protein Data Bank, as compared with over **200,000 protein structures**. We look forward to seeing improved experimental methods that will make RNA tertiary structure determination more routine, and to the development of advanced machine-learning approaches that may help to crack the RNA structure code.

Cell typing by electrophysiology

Extracellular recordings provide a wealth of information about neuronal activity but it is difficult to link this information to cell types.

By Nina Vogt

The advent of high-density probes has enabled the recording of neuronal activity during animal behavior in high throughput. But to understand neural circuit function, how behaviors arise and how the brain processes information, it is important to characterize the physiology, the connectivity, the morphology and the molecular identity of the neurons involved. Hence, units identified via extracellular recordings need to be linked to the underlying cell types.

Such a feat has been achieved by combining patch-clamp recordings of single neurons with backfilling and morphological characterization, or with Patch-seq (*Nat. Biotechnol.* **34**, 199–203; 2016). However, these approaches require recording from single neurons and are therefore inherently low in throughput.

As the morphological or molecular characterization of neurons recorded with high-density probes is not possible, alternative approaches are needed. This is where machine-learning approaches enter the stage. Classifiers can be trained to identify